

Question Number	Answer	Mark
7(a)(i)	The only correct answer is B 2 <i>A is incorrect because the nucleus and mitochondria contain DNA</i> <i>C is incorrect because the nucleus and mitochondria contain DNA</i> <i>D is incorrect because the nucleus and mitochondria contain DNA</i>	(1)

Question Number	Answer	Additional guidance	Mark
7(a)(ii)	A description that includes two of the following points: • (mitochondria carry out) aerobic respiration (1) • provide {ATP / energy} (1) • to move the flagellum (1)	Accept respiration Do not accept '{produce / make} energy' unqualified Accept tail	(2)

Question Number	Answer	Additional guidance	Mark
7(a)(iii)	A calculation showing the following steps: - conversion of actual length of scale into μm (1) - actual length divided by the given length (1)	<u>Example of calculation:</u> 40 mm = 40000 μm Allow answer in the range of 40000 to 41000 μm (40000 $\div$ 60) = 666.67 to 683.3 Correct answer with no working shown gains both marks Allow ECF	(2)

Question Number	Answer	Additional guidance	Mark
7(b)	An answer that includes the following points: - sperm (with longer flagellum) swim {fast / faster} (to the egg cell) (1) (sperm have) increased chance of fertilising (the egg cell) (1) - competition with sperm from other (male Chinese) hamsters (1)	Accept (longer flagellum) gives more {force / movement} Accept to be the first to fertilise the egg cell Ignore to be the first to reach the egg cell Accept description of competition with sperm from other (male Chinese) hamsters Accept as a consequence of natural selection / evolution	(3)

Question Number	Answer	Additional guidance	Mark
7(c)	An explanation that includes the following points: - cortical {granules / vesicles} fuse with (egg cell surface) membrane and release enzymes (1) zona pellucida hardens (1) - to prevent {polyspermy / more than one sperm (nucleus) entering egg cell} / to ensure that the nucleus is diploid (1)	Ignore chemicals Accept 'thickens' or 'becomes impermeable' Accept 'no other sperm can enter'	(3)

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Question Number	Answer	Additional guidance	Mark
7(d)	A description that includes three of the following points: • differential gene expression (1) • some genes have been (permanently) {inactivated / switched off} (1) • by epigenetic modification (1) • {proteins / enzymes} {made / synthesised} (from active genes) which permanently modify the cell (1)	Accept only some genes are {active / switched on} Ignore genes activated Accept named example of modification e.g. histone modification, DNA methylation Accept {proteins / enzymes} {made / synthesised} produce a {structural / functional / metabolic change}	(3) exp